

# Machine Learning Day 4

## Evaluation Metrics

Evaluation metrics are used to measure the performance of the machine learning models. Earlier in this introduction to machine learning, we saw that most problems are either regression and classification, and they are evaluated differently. Let us start with metrics that are used for evaluating regression problems

## Regression Metrics

In regression tasks, the goal is to predict the continuous value. The difference between the actual value and the predicted value is called the *error*.

*Error = Actual value - Predicted value*

The square of the error over all samples is called Mean Squarred Error(MSE).

*MSE = SQUARE(Actual value - Predicted value)/Number of Samples*

**MSE Actual Formula:**

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Taking the square root of the mean squared error will give the Root Mean Squared Error(RMSE). RMSE is the most used regression metric.

**RMSE Actual Formula:**

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

There are times that you will work with the datasets containing outliers. A suitable metric for those kinds of datasets is Mean Absolute Error (MAE). As simple as

calculating MSE, MAE is also the absolute of the error.

*MAE = ABSOLUTE (Actual value - Predicted Value)*

### **MAE Actual Formula**

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Like said, MAE is very sensitive to outliers. It is a suitable metric for all kinds of problems that are likely to have abnormal scenarios such as time series.

### **Classification Metrics**

In classification problems, the goal is to predict the categories/class.

*Accuracy* is the most commonly used metric. The accuracy shows the ability of the model in making the correct predictions. Take an example, in a horse and human classifier. If you have 250 training images for horses and the same number for humans, and the model can confidently predict 400 images, then the accuracy is  $400/500 = 0.8$ , so your model is 80% accurate.

The accuracy is simply an indicator of how your model is in making correct predictions and it will only be useful if you have a balanced dataset (like we had 250 images for horses and 250 images for humans).

When we have a skewed dataset or when there are imbalances, we need a different perspective on how we evaluate the model. Take an example, if we have 450 images for horses and 50 images for humans, there is a chance of 90% ( $450/500$ ) that the horse will be correctly predicted because the dataset is dominated by the horses. But how about humans? Well, it's obvious that the model will struggle at predicting them correctly.

This is where we introduce other metrics that can be far more useful than accuracy such as *precision*, *recall*, and *F1 score*.

*Precision* shows the percentage of the positive predictions that are actually positive.